PROJECT: TITLE : CROP RECOMMENDATION
SUBMITTED BY : Ajanya K A

TABLE OF CONTENTS

SL NO	CONTENTS	PAGE NO
1	ABSTRACT	3
2	INTRODUCTION	4
3	LITERATURE STUDY	5
4	ARCHITECTURE DIAGRAM	14
5	METHODOLOGY	15
6	TECHNICAL ARCHITECTURE	17
7	MACHINE LEARNING MODEL-RANDOM FOREST CLASSIFIER 9.1. INTRODUCTION 9.2. WORKING MECHANISM 9.3. FEATURE RANDOMNESS 9.4. DECISION MAKING 9.5. FEATURE IMPORTANCE AND INTERPRETABILITY 9.6. ADVANTAGES 9.7. IN SMART CROP RECOMMENDATION	18
8	IMPLEMENTATION-CODING	20
9	RESULTS AND ANALYSIS	28
10	CONCLUSION AND FUTURE ENHANCEMENT	29

1. ABSTRACT

The **Smart Farming Crop Recommendation System** is a web-based application designed to assist farmers in selecting the most suitable crops based on soil and environmental conditions. Using a **Random Forest machine learning model**, it analyses inputs like **Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall** to suggest optimal crops, helping to increase yield and reduce risks.

The system also features a **dashboard with over 10 interactive charts** for visual insights into soil and weather patterns. A **chatbot assistant** provides instant answers to farming-related queries such as fertilizer use, pest control, and smart irrigation practices.

Built with **Flask, Python, HTML, CSS, and JavaScript**, this tool integrates prediction, visualization, and interaction in one interface, promoting data-driven decisions and supporting smart agriculture in both rural and modern settings.

2. INTRODUCTION

Agriculture plays a pivotal role in sustaining the global economy and ensuring food security. However, modern-day farmers face numerous challenges, such as unpredictable weather conditions, declining soil fertility, water scarcity, and the increasing demand for higher crop productivity. To overcome these hurdles and make farming more efficient, the integration of Artificial Intelligence (AI) and Machine Learning (ML) into agriculture has emerged as a transformative solution.

This project, titled "Smart Farming Crop Recommendation System", aims to empower farmers by providing them with accurate and data-driven crop suggestions. By analysing crucial parameters like soil nutrients (Nitrogen, Phosphorus, Potassium), weather conditions (temperature, humidity, rainfall), and soil pH, the system predicts the most suitable crop to grow in a particular region. This helps optimize land use, improve yield, and reduce the guesswork involved in traditional farming decisions.

The project leverages a Random Forest algorithm, a robust ML model known for its high accuracy and ability to handle nonlinear data. Additionally, the system includes a data visualization dashboard with interactive charts to help users understand their data better, as well as a real-time chatbot assistant with speech-to-text functionality to provide farmers with instant, voice-driven support on agricultural best practices.

By combining predictive analytics, visual insights, and intelligent conversation, the Smart Farming Crop Recommendation System acts as a virtual agricultural advisor. It serves as a step towards precision agriculture, where data and technology come together to make farming smarter, more sustainable, and more profitable.

3. LITERATURE SURVEY

3.1. Crop Recommendation System Using Machine Learning – Rajak et al. (2017)

Rajak et al. (2017) proposed a crop recommendation system designed to assist farmers in selecting suitable crops based on soil characteristics. The system leverages machine learning algorithms to enhance the precision of crop recommendations, focusing on four major crops: rice, cotton, sugarcane, and wheat. The core idea revolves around using an ensemble learning technique, combining three distinct classification models:

- Random Forest
- Naive Bayes
- Support Vector Machine (SVM)

To improve prediction reliability, the authors implemented a majority voting mechanism where the final recommendation is made based on the consensus of the classifiers. The system uses a pre-processed soil dataset and classifies crop types according to features extracted from that dataset.

The model aims to maximize crop yield by recommending the most appropriate crop based on patterns found in historical data. It relies on soil properties such as nutrient composition and other relevant agricultural attributes.

Advantages

- High Prediction Accuracy Through Ensemble Learning:
  By combining the strengths of multiple algorithms, the system achieves greater predictive performance than using a single classifier. This ensemble approach minimizes overfitting and improves generalization on unseen data.
- Flexibility Across Multiple Crop Types:
  The model supports classification for a variety of crops, making it adaptable to different regions and conditions, provided the training dataset is comprehensive.
- Scalability:
  Since the base models are relatively lightweight, the system can be scaled and adapted for different soil datasets or expanded with additional classifiers if needed.
- Ease of Integration:

The structure allows it to be integrated into web or mobile-based systems to aid farmers directly in the field.

Disadvantages

- Lack of Real-Time Soil Condition Analysis:
  The model assumes soil parameters based on historical datasets, meaning it does not validate or assess current real-time soil conditions. This limits the system's responsiveness to sudden environmental changes or soil degradation.
- Limited Crop Diversity:
  Although effective, the system only focuses on a small group of crops (rice, cotton, sugarcane, and wheat), which might not cater to the needs of all regions or farming contexts.
- No Environmental or Weather Integration:
  Factors like weather conditions, rainfall, temperature, or pest outbreaks are not considered in the decision-making process, reducing the system's robustness for real-world deployment.
- Majority Voting Could Mask Best Individual Predictions:
  While ensemble learning is beneficial, in certain cases, the majority voting system could lead to misclassification if the stronger model is outvoted by weaker ones.

Conclusion

The system proposed by Rajak et al. provides a strong foundation for a data-driven crop recommendation system using ensemble machine learning methods. It shows the potential of machine learning to assist in agricultural decision-making. However, for practical, real-time farming applications, the model needs to incorporate live soil sensor data, weather forecasts, and pest/disease analysis to become truly intelligent and responsive. The study highlights the importance of integrating multiple algorithms and sets the stage for future improvements in precision agriculture systems.

3.2. Crop Recommendation System for Precision Agriculture – Pudumalar et al. (2017)

Pudumalar et al. (2017) presented a **precision agriculture-based crop recommendation system** designed to provide accurate and site-specific crop suggestions. Their model focuses on utilizing **soil characteristics, crop yield history, and environmental parameters** to determine the most suitable crop for a specific location. This approach is a move away from general recommendations and leans toward **data-driven, site-specific decision-making** in farming.

The authors employed a **machine learning ensemble model** combining three algorithms:

- **Random Forest**
- **K-Nearest Neighbors (KNN)**
- **Naive Bayes**

A **majority voting technique** was used to decide the final recommendation, improving the robustness of the system. The goal was to minimize incorrect crop selection and optimize productivity through accurate data analysis.

This system works by collecting **soil test data and climatic parameters**, analyzing them, and then providing personalized crop suggestions for that specific farm or region.

Advantages

- **Ensemble Model with High Accuracy:**
  The combination of Random Forest, KNN, and Naive Bayes ensures more reliable and accurate crop recommendations through majority voting.
- **Site-Specific Recommendations:**
  Instead of relying on generalized data, the system uses **site-specific parameters** (like soil type, moisture, fertility, etc.) to recommend crops, which makes it highly useful for precision farming.
- **Minimizes Resource Misallocation:**
  By suggesting the right crop for the right location, it helps avoid unnecessary usage of fertilizers, water, and land, promoting **efficient resource utilization**.
- **Boosts Productivity and Profitability:**

Accurate recommendations based on detailed site analysis can result in **higher crop yields**, better income for farmers, and reduced risk of crop failure.

Disadvantages

- **Limited Dynamic Adaptability:**
  Although the system considers site-specific inputs, it may **not respond dynamically** to changing weather conditions or pest outbreaks during the crop cycle.
- **No Pest or Disease Analysis:**
  The system doesn't incorporate pest or disease forecasting, which are **crucial factors** in determining the success of a particular crop.
- **Requires Quality Data Inputs:**
  The accuracy of the recommendation depends heavily on the **quality and completeness of the input data**. In rural areas, acquiring accurate soil and environmental data may be challenging.
- **Technical Complexity for Farmers:**
  The system may require technical knowledge or tools (e.g., mobile apps, data input formats) that might be difficult for some farmers to use without assistance.

Conclusion

The study by Pudumalar et al. represents a **significant step forward in applying precision agriculture principles** to real-world farming challenges. By leveraging an ensemble of machine learning models, the system provides **customized crop recommendations** that can enhance productivity while promoting sustainable land use. However, for broader and more effective implementation, the model should be integrated with **real-time data feeds**, **pest/disease detection modules**, and a more **user-friendly interface** tailored to non-technical users. The research highlights the **power of machine learning in transforming traditional agriculture** into a smarter and more efficient practice.

3.3. Recommendation System for Crop Identification and Pest Control – Kumar et al. (2019)

Kumar et al. (2019) developed a dual-purpose agricultural recommendation system that not only suggests the **most suitable crop** based on soil and climatic parameters but also **identifies pests** and provides corresponding **control techniques**. This integrated model addresses a critical gap in farming, where **lack of awareness about crop-specific needs and pest threats** can result in reduced yield and economic losses.

To tackle this problem, the authors implemented and compared three machine learning classification algorithms:

- **Support Vector Machine (SVM)**
- **Decision Tree**
- **Logistic Regression**

The dataset included essential agricultural features such as **soil moisture, nutrient levels, and environmental conditions**. Among the models tested, **SVM demonstrated superior performance** in terms of accuracy, making it the preferred choice for their system.

The recommendation system works in two stages:

- **Crop Prediction Module** – Predicts the most suitable crop for the given input parameters.
- **Pest Detection and Control Module** – Identifies potential pests and provides preventive or remedial pest control strategies.

Advantages

- **Dual Functionality (Crop + Pest Recommendations):**
  Unlike traditional systems, this model goes beyond just crop selection by also identifying pests and offering solutions, thereby making it **more comprehensive and practical for farmers**.
- **High Accuracy with SVM:**
  The Support Vector Machine algorithm outperformed others in prediction accuracy, making the system reliable for real-world agricultural decisions.
- **Addresses Farmers' Knowledge Gaps:**

Recognizes that farmers often lack complete information about soil requirements and pest threats, and fills that gap using **machine-driven insights**.

- **Prevention-Oriented Design:**
  The pest control recommendations are designed to **prevent crop damage before it occurs**, thereby **reducing financial losses and increasing yield**.

Disadvantages

- **Requires Accurate Environmental Data:**

  The system's effectiveness depends heavily on the availability of **real-time and accurate data** regarding soil and climatic conditions, which may be a limitation in rural or remote areas.

- **No Real-Time Monitoring:**

  The model is trained on **static historical data** and doesn't integrate real-time sensor-based updates or weather forecasting.

- **Potential Complexity for Small Farmers:**

  Users may need assistance to operate or interpret the recommendations if presented through a technical interface or software.

- **Lacks Ensemble or Hybrid Modeling:**

  Although multiple algorithms are tested, the final implementation is based on a single best-performing model (SVM), missing the additional robustness that **ensemble learning could offer**.

Conclusion

Kumar et al. (2019) introduced a **multi-faceted agricultural recommendation system** that enhances the traditional scope of crop suggestion by integrating **pest detection and management**. This makes the system more aligned with real-world farming challenges where both crop suitability and pest threats must be managed simultaneously. With a strong foundation in machine learning, especially **SVM**, the system holds promise for increasing productivity and reducing farmer distress. However, further enhancements like **real-time data integration**, **IoT-based sensing**, and **user-friendly mobile interfaces** could make it more impactful and widely adoptable.

3.4. A Study on Various Data Mining Techniques for Crop Yield Prediction – Gandge et al. (2019)

Gandge et al. (2019) conducted a comprehensive study on using different data mining and machine learning techniques to predict crop yields across various crop types. The research considered important agronomic variables such as soil quality, NPK (Nitrogen, Phosphorus, Potassium) levels, and moisture content over time. Rather than relying on a single algorithm, the study tailored the model selection based on the crop type, demonstrating a multi-model prediction strategy.

They applied:

- Multiple Linear Regression (MLR) for rice crop yield prediction.
- K-Nearest Neighbors (KNN) for wheat and potato yield.
- Neural Networks for corn crop prediction.

This crop-specific modeling showcases that a tailored algorithm often performs better than using a single method universally.

Advantages

- Tailored Modeling Approach: Algorithms are chosen based on specific crop data behavior, improving prediction accuracy per crop.
- Variety of Algorithms Explored: A mix of regression and classification algorithms shows versatility and adaptability.
- Comprehensive Feature Usage: Soil nutrients, moisture levels, and other environmental factors were all used as input features.
- Good Learning Resource: Ideal for benchmarking and understanding algorithm performance differences.

Disadvantages

- No Unified Crop Recommendation System: Each crop uses a separate algorithm; there's no integrated platform or dynamic model selection mechanism.
- Does Not Recommend What to Grow: The focus is strictly on yield prediction, not on suggesting the most suitable crop.
- No Real-Time Data Integration: Lacks sensor-based or dynamic environmental inputs for live decision-making.
- Limited Evaluation Metrics: The study does not provide detailed performance comparisons (e.g., accuracy, RMSE) for each algorithm.

Conclusion

Gandge et al.'s work provides valuable insights into how different machine learning algorithms perform on crop-specific yield prediction. While highly relevant from an academic and research perspective, the absence of a unified system and real-time adaptability makes it less practical for immediate field deployment. Future improvements could involve integrating all models into a smart platform that includes crop recommendation features and live data feeds via IoT.

3.5. Research on Precision Management of Farming Season Based on Big Data – Yan et al. (2018)

Yan et al. (2018) focused on precision management of farming seasons using meteorological big data. Instead of crop recommendation, this work aims to determine the optimal time to start farming activities. The authors used thirteen weather-related features including atmospheric pressure, temperature, light, and precipitation, then narrowed down to three key features after discretization:

- Accumulated Temperature
- Sunshine Hours
- Temperature

They applied the Apriori algorithm, an association rule mining technique, to generate rules and determine ideal sowing months based on climatic data. This system helps farmers align their crop planting schedules with the best possible seasonal window for optimal yield.

Advantages

- Focus on Seasonal Optimization: Helps farmers identify the most suitable time to begin planting, which is a critical factor in agricultural success.
- Use of Meteorological Big Data: Demonstrates how large datasets can be transformed into actionable rules using data mining.
- Efficient Rule-Based Model: Apriori is lightweight, interpretable, and easy to implement in decision support systems.

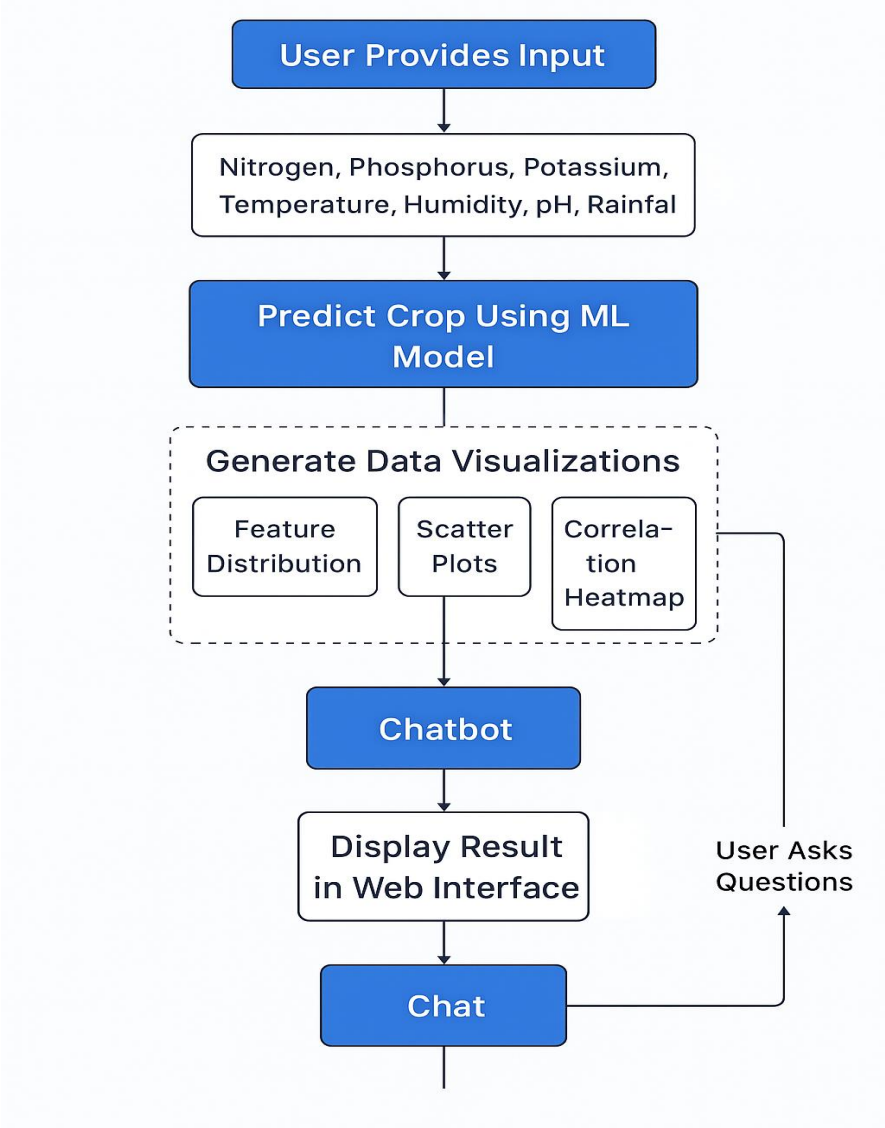
Disadvantages

- Does Not Recommend Crop Type: The system only advises when to sow, not what to sow, limiting its utility for crop selection.
- Feature Reduction May Impact Accuracy: Reducing features to only three may cause loss of predictive power in certain contexts.
- No Integration of Soil Data or Yield Metrics: The model focuses only on weather; soil fertility, crop history, and resource constraints are not considered.
- Static Association Rules: Apriori produces fixed rules, which may not adapt to real-time climate variability or abnormal weather patterns.

Conclusion

The study by Yan et al. (2018) is an excellent example of how weather data can be mined for seasonal planning in agriculture. While it doesn't address crop selection, its ability to help farmers decide when to plant can significantly impact productivity. However, for broader usability, this system should be integrated with crop recommendation logic, soil conditions, and real-time weather updates. A hybrid approach combining seasonal timing and crop selection would provide a more complete precision agriculture solution.

4. ARCHITECTURE DIAGRAM



5. METHODOLOGY

5.1. Requirement Analysis:

The project began with a comprehensive requirement analysis to understand the fundamental issues faced by farmers. Through initial research, it was identified that farmers often struggle to choose the right crop due to limited knowledge of soil properties, weather conditions, and resource optimization. The aim of the system was set to provide accurate, data-driven crop recommendations by integrating machine learning and a user-friendly web application. These insights helped shape the scope, goals, and features of the Smart Farming Crop Recommendation System.

5.2. Data Collection:

Once the problem was well-defined, the next step involved sourcing a reliable and relevant dataset. A publicly available Crop Recommendation dataset was selected, containing vital features such as nitrogen, phosphorus, potassium, temperature, humidity, pH level, and rainfall. This dataset was chosen for its completeness and alignment with the requirements of building a predictive model for agriculture. Ensuring high data quality was prioritized to avoid introducing biases or inaccuracies into the prediction system.

5.3. Data Preprocessing:

To prepare the dataset for model training, several preprocessing steps were performed. Missing values were handled, and all features were normalized where required to bring them to a similar scale. The label column (crop names) was encoded using Label Encoding to convert them into numerical values suitable for machine learning models. The dataset was then split into training and testing sets in an 80:20 ratio to allow for performance evaluation after model training.

5.4. Model Training:

The machine learning model was developed using a Random Forest Classifier due to its efficiency in handling large datasets and its ability to capture complex relationships among features. The model was trained on the preprocessed dataset and evaluated on the testing data. It showed strong accuracy and generalization capabilities. After successful training, the model

and label encoder were saved using Joblib for later use during prediction in the web application.

5.5. **Web Application Development:**

With the model ready, the focus shifted to building a web interface using Flask. The interface allows users to input values such as soil nutrients and weather parameters, which are then passed to the trained model for prediction. The backend retrieves the predicted crop and displays the result on the webpage. A clean and simple UI was created using HTML and CSS, ensuring the application is accessible and usable by individuals with minimal technical experience.

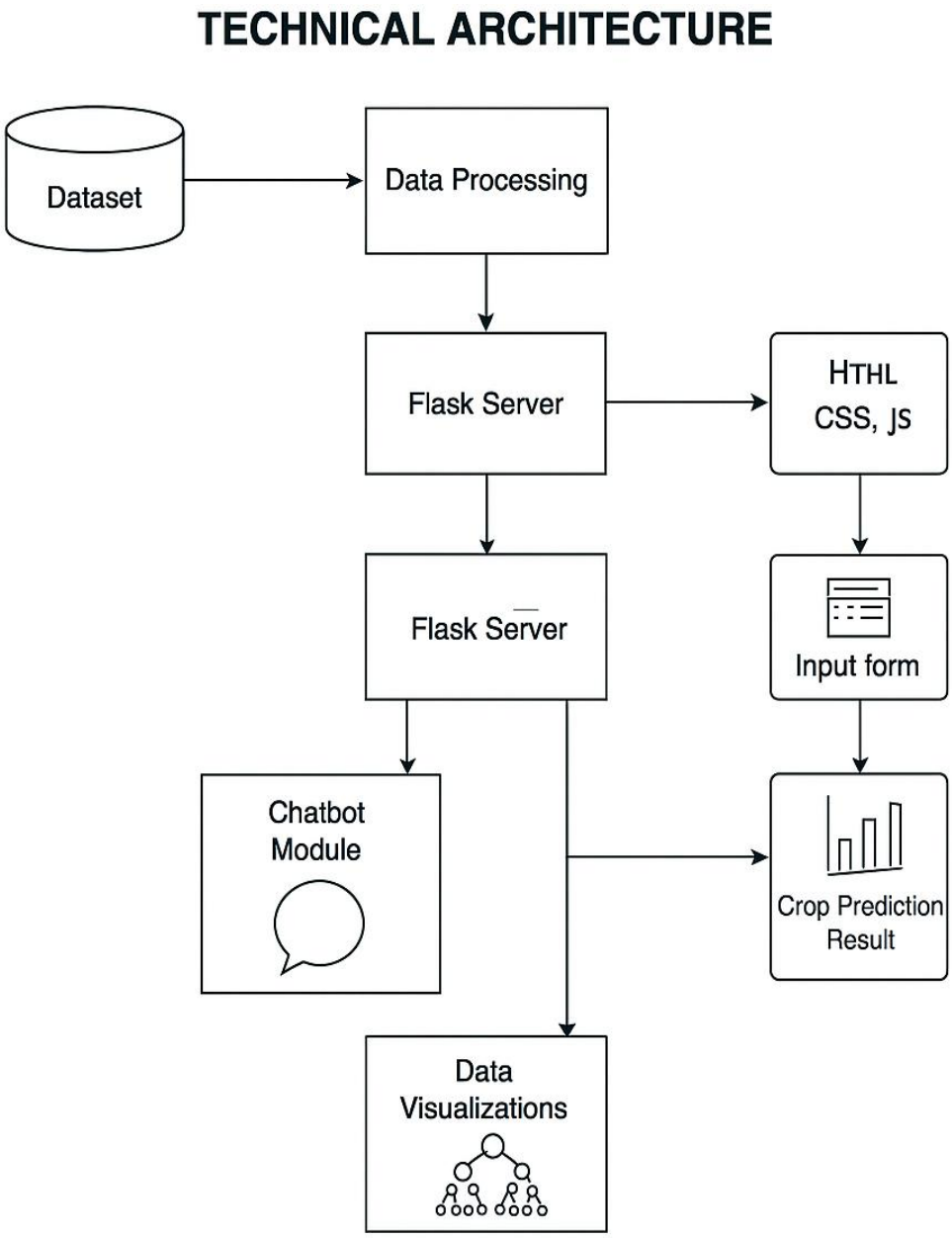
5.6. **Chatbot Integration:**

To improve user engagement and provide guidance, a simple rule-based chatbot was integrated into the web interface. The chatbot responds to predefined farming-related queries using a logic-based script in JavaScript and Flask. It operates in real time using AJAX to avoid page reloads. While not based on natural language processing or AI, this chatbot provides basic assistance such as crop tips, fertilizer suggestions, and general farming knowledge.

5.7. **Visualization and Testing:**

The final step involved adding visual analytics and conducting comprehensive testing. Several charts were created using Matplotlib to display feature importance, nutrient distributions, and other useful insights for users. These visualizations are shown on the web interface to aid understanding. The application was tested thoroughly to ensure smooth functionality of the prediction system, chatbot interaction, and chart rendering. Edge cases and incorrect inputs were also tested to validate the robustness of the system.

6. TECHNICAL ARCHITECTURE



7. MACHINE LEARNING MODEL-RANDOM FOREST

7.1. Introduction to Random Forest

The Random Forest Classifier is a supervised machine learning algorithm that is widely used for both classification and regression problems. It is an ensemble learning method, which means it builds multiple decision trees and merges them together to get a more accurate and stable prediction. The basic idea is to create a 'forest' of decision trees, each trained on a different subset of the data, and to make predictions based on the majority vote of these trees. This approach helps to improve the performance and reliability of predictions compared to a single decision tree.

7.2. Working Mechanism

Random Forest works by generating multiple decision trees during the training phase. For each tree, it uses a different subset of the training data, selected randomly with replacement—a method known as bootstrap sampling. Additionally, at each node in a tree, it randomly selects a subset of features to determine the best split. This introduces diversity among the trees, ensuring that they are not all biased toward the same patterns in the data. When making predictions, each tree gives a classification result, and the Random Forest model chooses the result with the most votes from all the trees.

7.3. Feature Randomness and Data Sampling

To make the model robust, Random Forest incorporates two layers of randomness: one in the data and one in the feature selection. The data sampling ensures that each tree is trained on a different view of the dataset, while the feature sampling ensures that the model does not rely too heavily on any single attribute. This combination of randomness improves generalization and reduces the chance of overfitting, which is a common issue with decision trees.

7.4. Decision Making in Classification

During the prediction stage, the Random Forest Classifier aggregates the decisions of all the individual trees. Each tree independently predicts the class of a data point, and the final class is chosen based on majority voting. For example, if 70 trees out of 100 classify a sample as "wheat" and 30 trees classify it as "rice",

the final prediction will be "wheat". This majority voting method makes the prediction more reliable and less sensitive to errors from individual trees.

7.5. Feature Importance and Interpretability

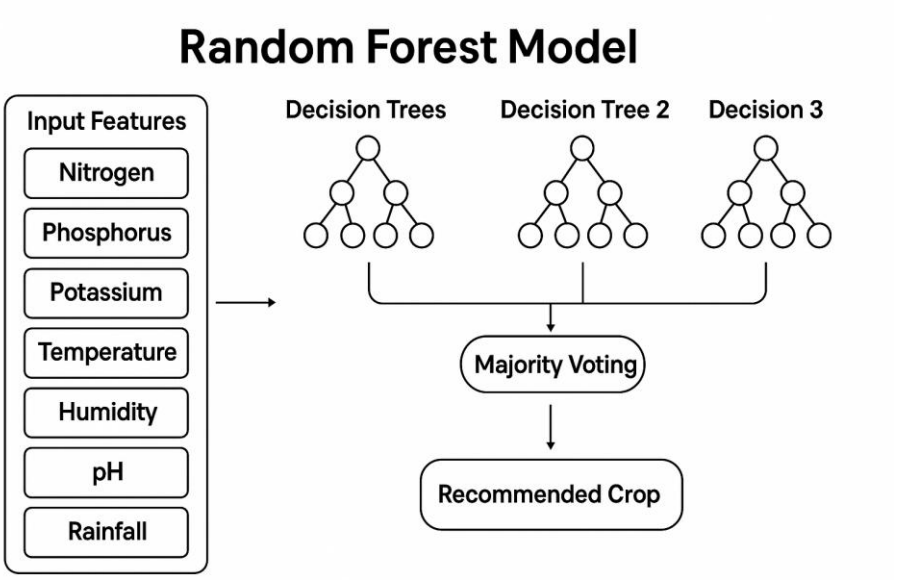
One of the useful outputs of a Random Forest model is the importance score of each feature. This indicates how useful or valuable each feature is in the construction of the decision trees. For instance, in a crop recommendation system, features such as soil pH, nitrogen level, and rainfall might have higher importance compared to temperature or potassium, depending on their influence on the crop type. Understanding feature importance can guide users or researchers in refining their data collection and analysis.

7.6. Advantages of Random Forest

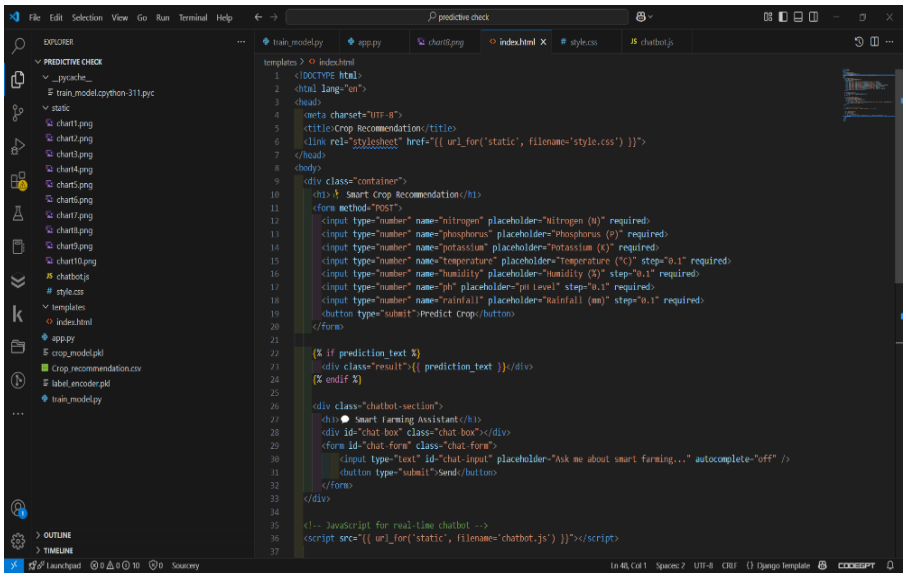
Random Forest is favoured for its high accuracy, robustness to noise, and minimal need for parameter tuning. It performs well on large datasets and can handle thousands of input features without feature deletion. Moreover, it is less likely to overfit than a single decision tree, especially when tuned with a sufficient number of trees. These properties make it a preferred choice in real-world applications, such as medical diagnosis, financial forecasting, and agricultural systems like smart crop recommendations.

7.7. Use in Smart Farming Crop Recommendation

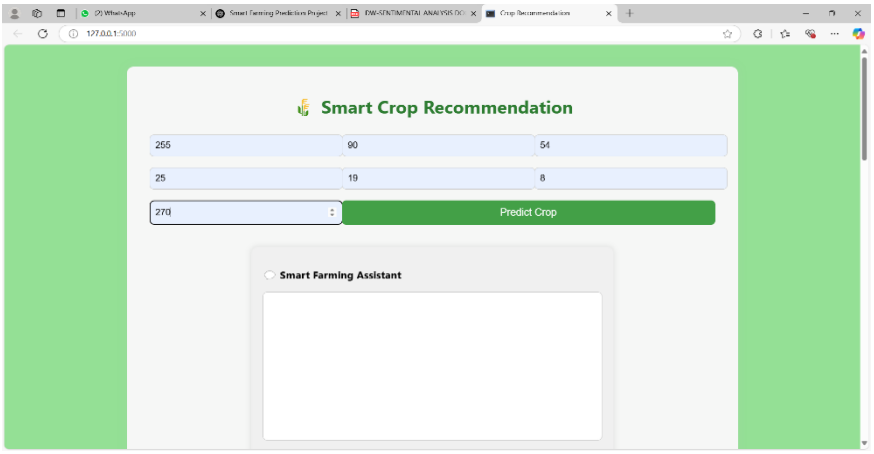
In the context of smart farming, Random Forest plays a crucial role in predicting the most suitable crop based on various environmental and soil factors. It processes inputs such as nitrogen, phosphorus, potassium levels, temperature, humidity, pH, and rainfall to suggest the best crop for a given land condition. Its accuracy and resilience to data inconsistencies ensure that farmers receive reliable recommendations. This contributes significantly to informed agricultural decision-making, optimizing resource usage and maximizing yield.

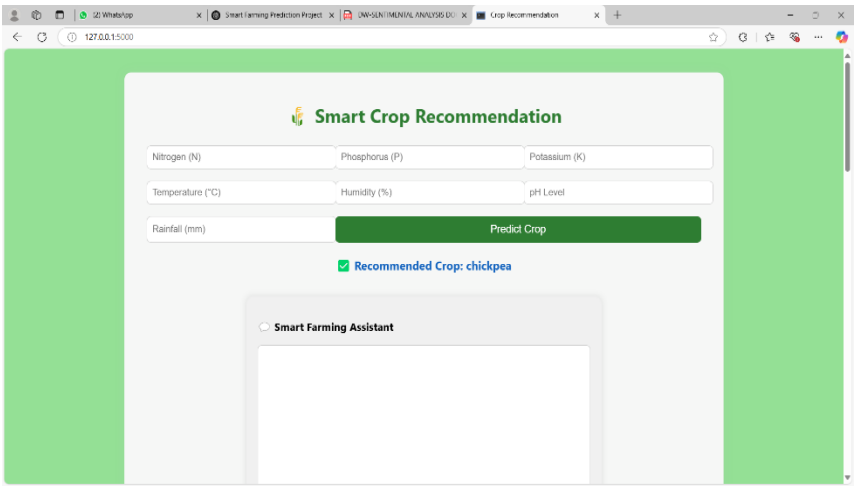


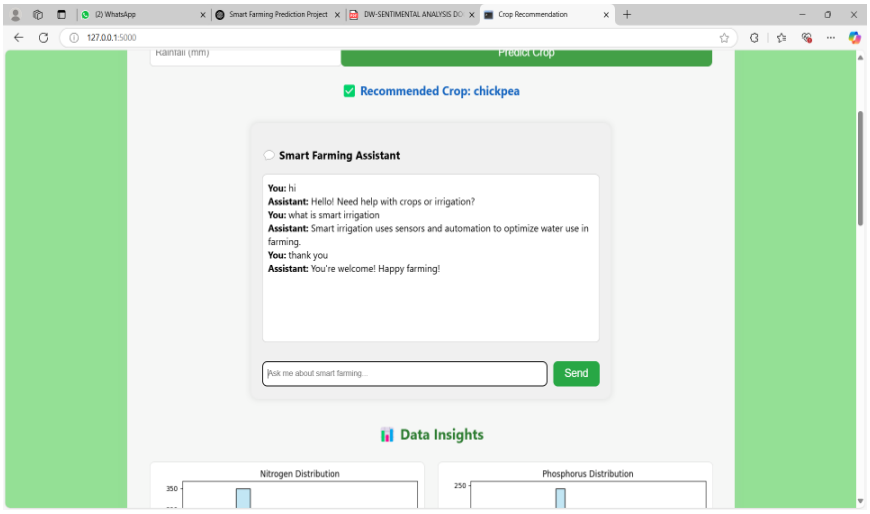
8. IMPLEMENTATION: CODING



OUTPUT:

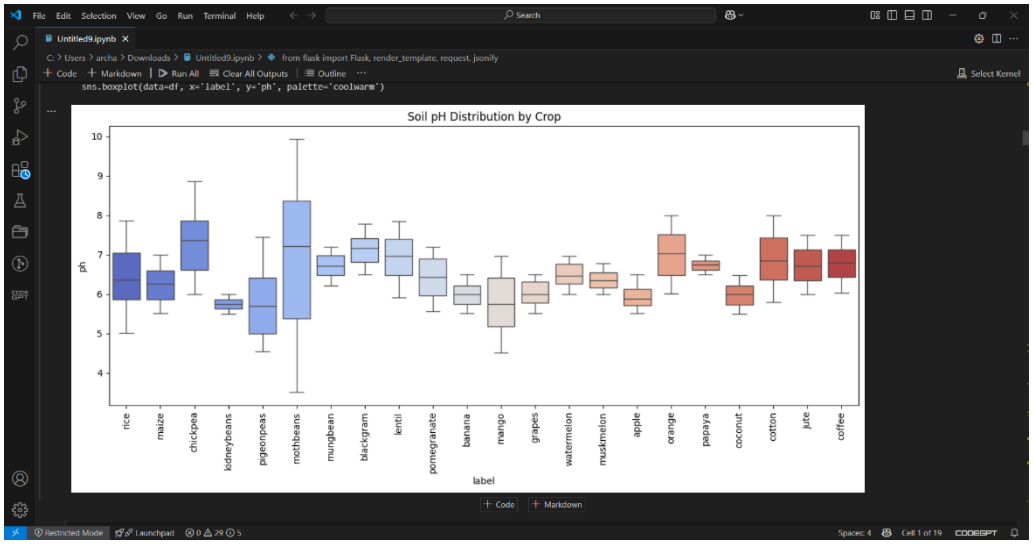

Smart Crop Recommendation
255
90
54
25
19
8
270
Predict Crop
Smart Farming Assistant

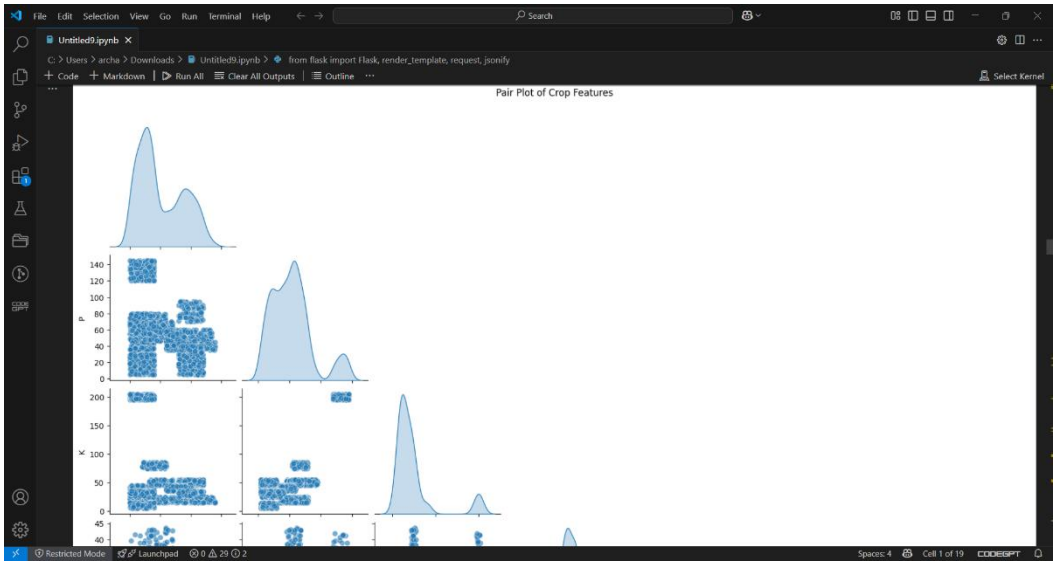

Smart Crop Recommendation
Nitrogen (N)
Phosphorus (P)
Potassium (K)
Temperature (°C)
Humidity (%)
pH Level
Rainfall (mm)
Predict Crop
Recommended Crop: chickpea
Smart Farming Assistant

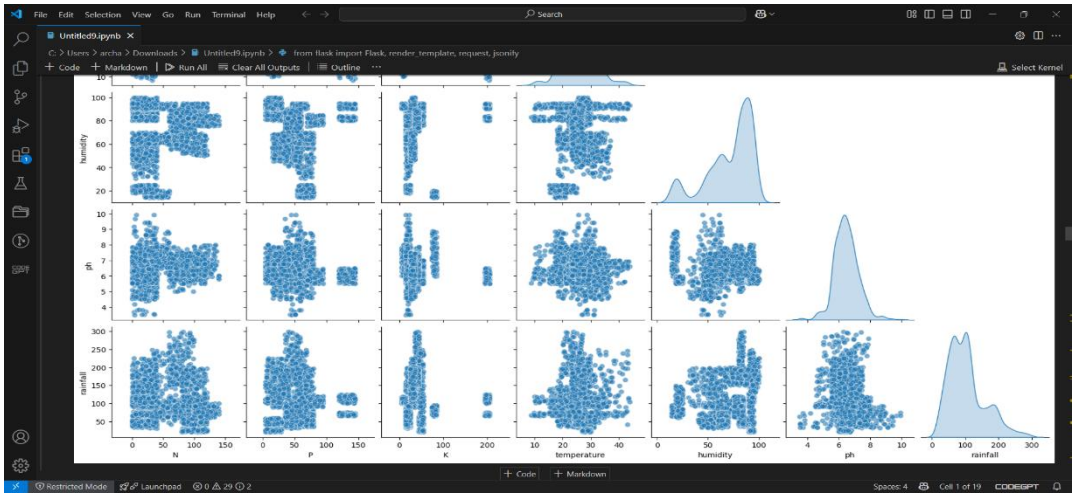

Rainfall (mm)
Predict Crop
Recommended Crop: chickpea
Smart Farming Assistant
You: hi
Assistant: Hello! Need help with crops or irrigation?
You: what is smart irrigation
Assistant: Smart irrigation uses sensors and automation to optimize water use in farming.
You: thank you
Assistant: You're welcome! Happy farming!
Ask me about smart farming...
Send
Data Insights
Nitrogen Distribution
Phosphorus Distribution

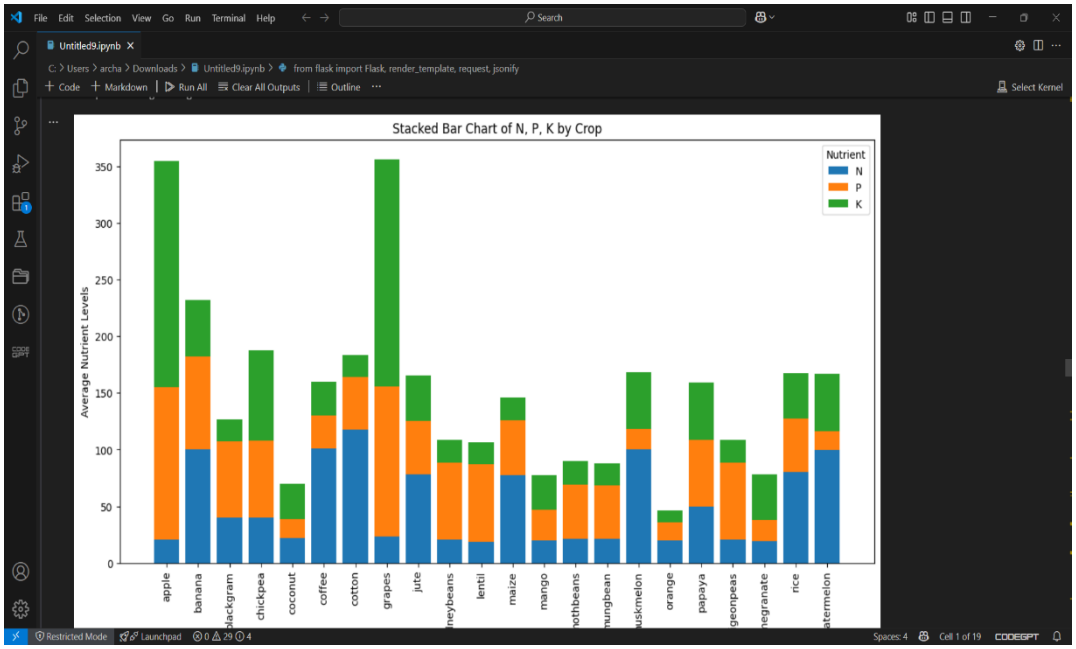

Temperature vs Humidity by Crop
Humidity (%)
Temperature (°C)
Crop
rice
maize
chickpea
kidneybeans
pigeonpeas
mothbeans
mungbean
blackgram
lentil
pomegranate
banana
mango
grapes
watermelon
muskmelon
apple
orange
papaya
coconut
cotton
jute
coffee

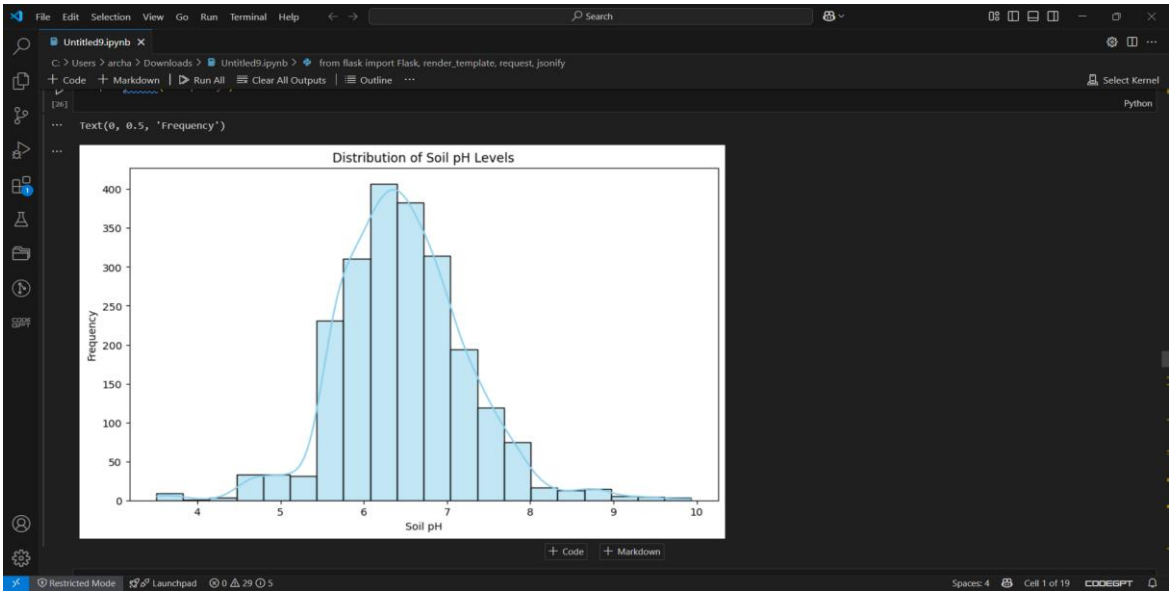

Correlation Heatmap of Soil & Weather Parameters
N
P
K
temperature
humidity
ph
rainfall

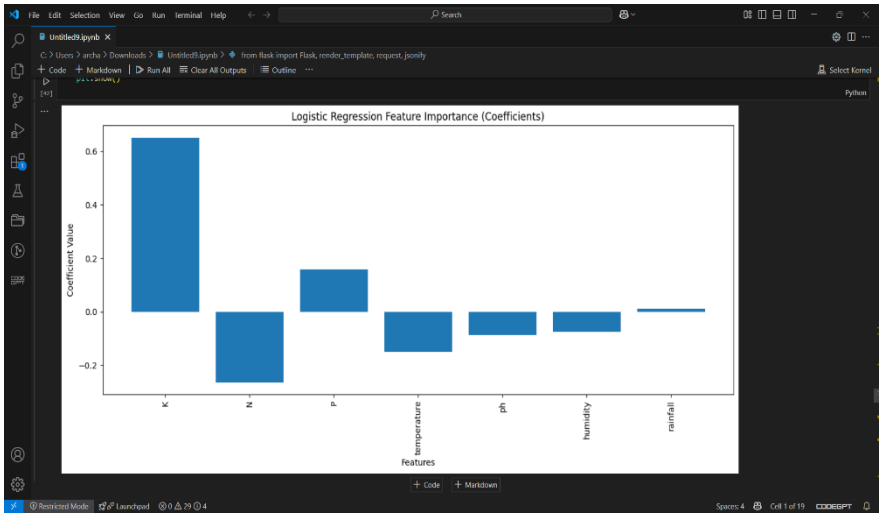

Soil pH Distribution by Crop
ph
label
rice
maize
chickpea
kidneybeans
pigeonpeas
mothbeans
mungbean
blackgram
lentil
pomegranate
banana
mango
grapes
watermelon
muskmelon
apple
orange
papaya
coconut
cotton
jute
coffee

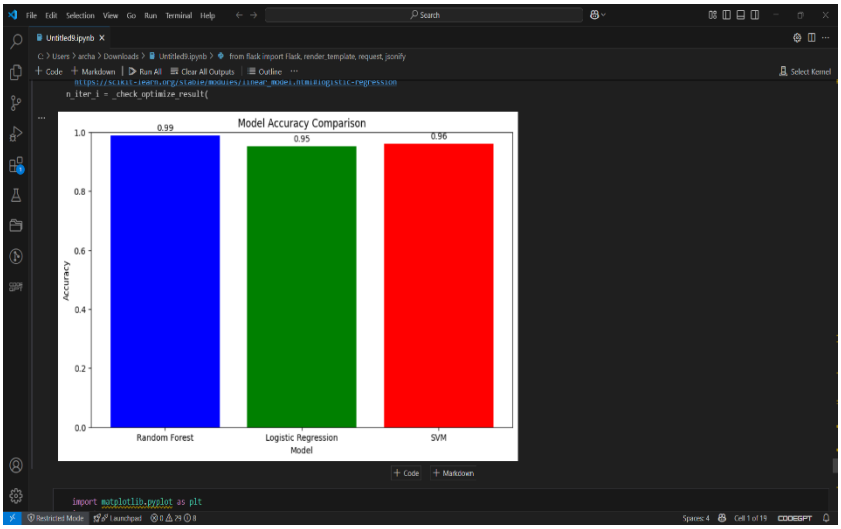

Pair Plot of Crop Features
P
K

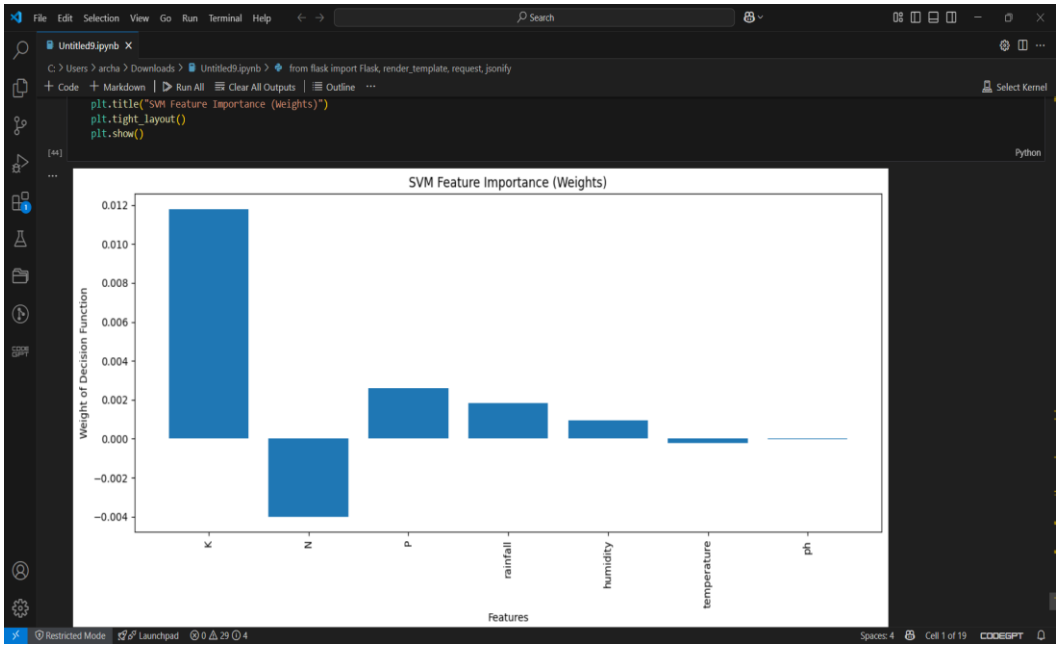

humidity
ph
rainfall
N
P
K
temperature
humidity
ph
rainfall

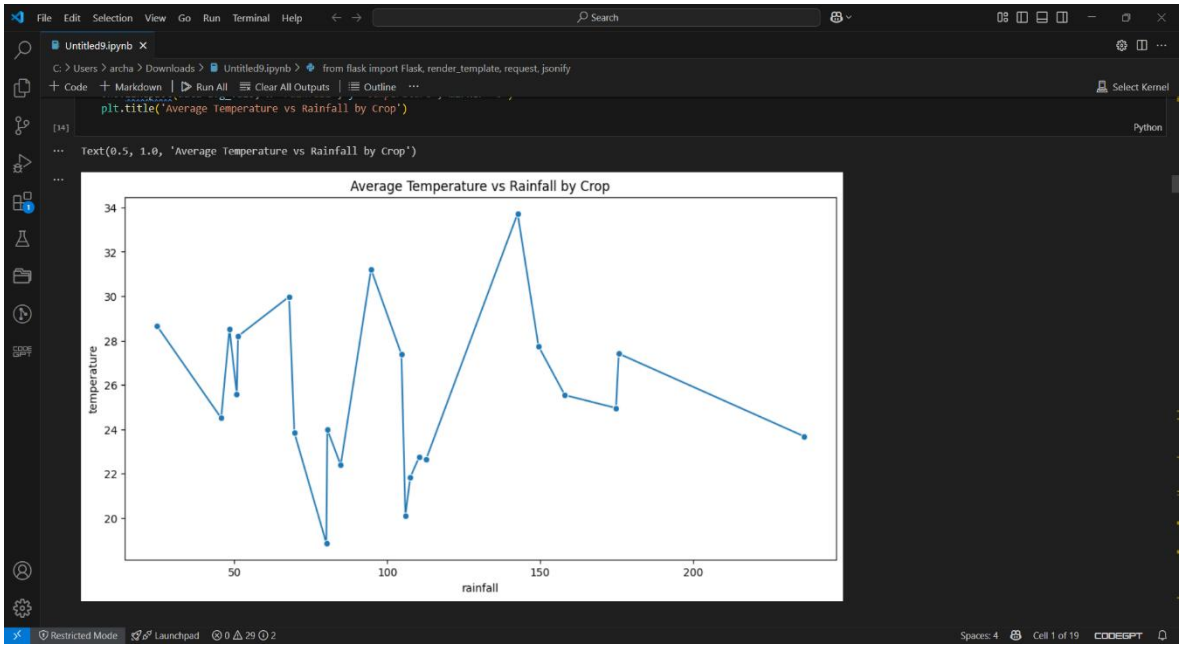

Stacked Bar Chart of N, P, K by Crop
Nutrient
N
P
K
Average Nutrient Levels
apple
banana
blackgram
chickpea
coconut
coffee
cotton
grapes
jute
lentil
maize
mango
papaya
rice
watermelon

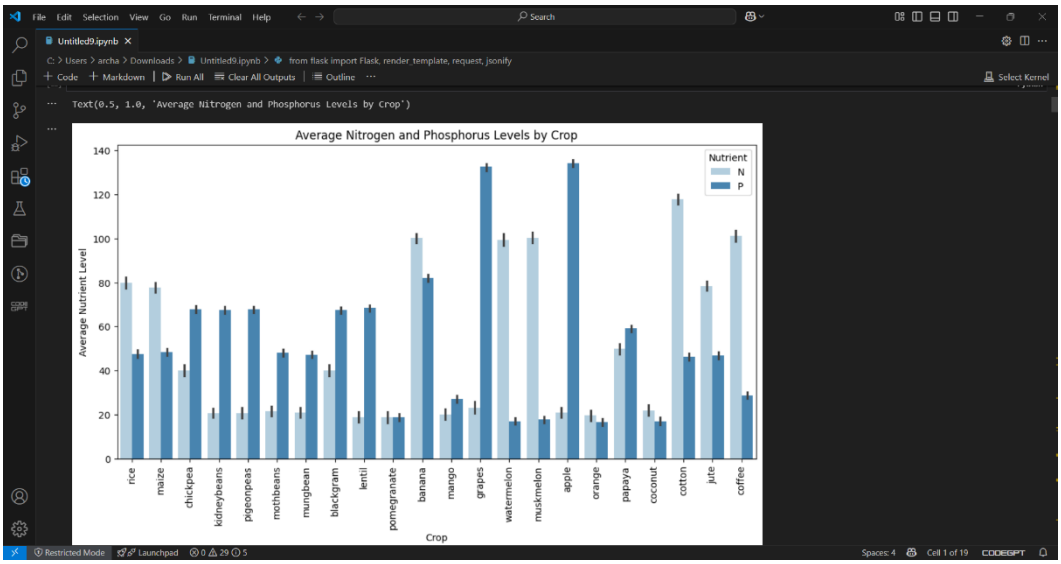

Distribution of Soil pH Levels
Frequency
Soil pH
Text(0, 0.5, 'Frequency')

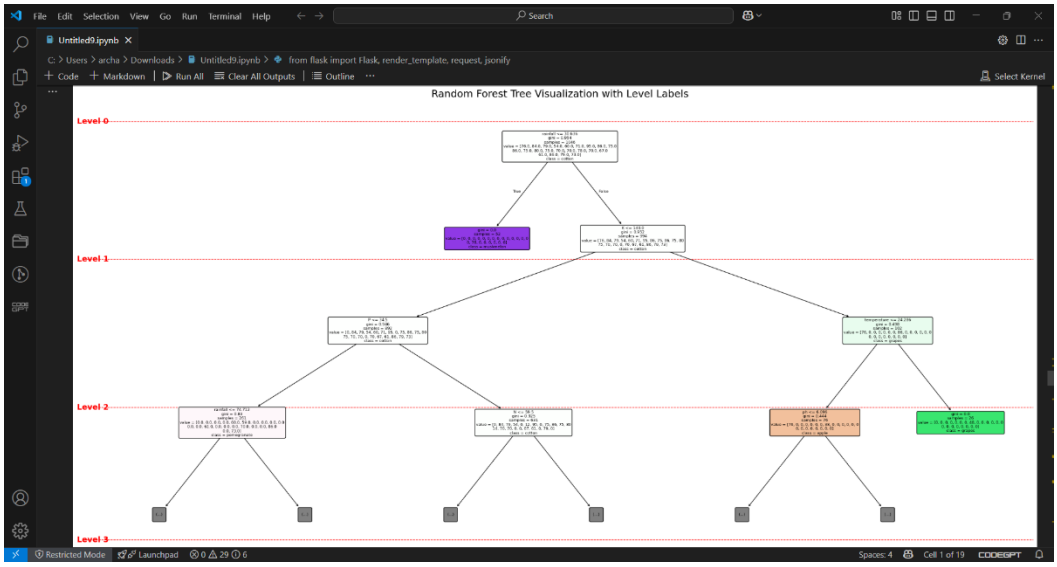

Logistic Regression Feature Importance (Coefficients)
Coefficient Value
Features
K
N
P
temperature
ph
humidity
rainfall

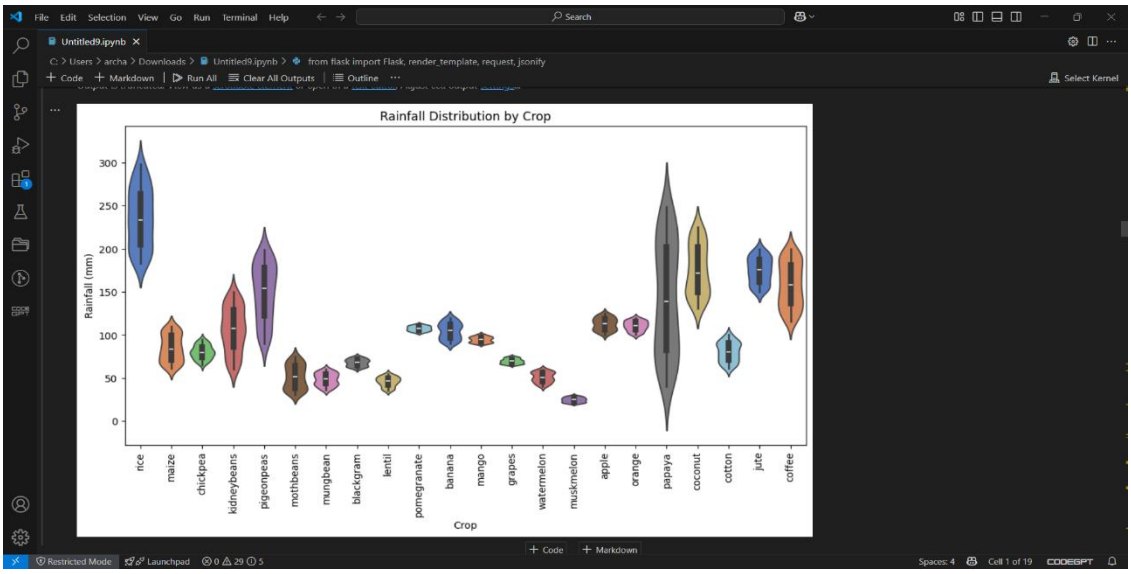

Model Accuracy Comparison
Accuracy
Random Forest
Logistic Regression
SVM
Model
0.99
0.95
0.96


File Edit Selection View Go Run Terminal Help
Search
Untitled9.ipynb
C: > Users > archa > Downloads > Untitled9.ipynb > from flask import Flask, render_template, request, jsonify
+ Code + Markdown | Run All Clear All Outputs | Outline
Select Kernel
plt.title("SVM Feature Importance (Weights)")
plt.tight_layout()
plt.show()
[44]
Python
SVM Feature Importance (Weights)
Weight of Decision Function
0.012
0.010
0.008
0.006
0.004
0.002
0.000
−0.002
−0.004
K
N
P
rainfall
humidity
temperature
ph
Features
Restricted Mode
Launchpad
Spaces: 4
Cell 1 of 19
CODEGPT


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Search
Untitled9.ipynb
C: > Users > archa > Downloads > Untitled9.ipynb > from flask import Flask, render_template, request, jsonify
+ Code + Markdown | Run All Clear All Outputs | Outline
Select Kernel
plt.title('Average Temperature vs Rainfall by Crop')
[14]
Python
Text(0.5, 1.0, 'Average Temperature vs Rainfall by Crop')
Average Temperature vs Rainfall by Crop
temperature
34
32
30
28
26
24
22
20
50
100
150
200
rainfall
Restricted Mode
Launchpad
Spaces: 4
Cell 1 of 19
CODEGPT


Untitled9.ipynb
Text(0.5, 1.0, 'Average Nitrogen and Phosphorus Levels by Crop')
Average Nitrogen and Phosphorus Levels by Crop
Nutrient
N
P
Average Nutrient Level
Crop
rice
maize
chickpea
kidneybeans
pigeonpeas
mothbeans
mungbean
blackgram
lentil
pomegranate
banana
mango
grapes
watermelon
muskmelon
apple
orange
papaya
coconut
cotton
jute
coffee


Untitled9.ipynb
Random Forest Tree Visualization with Level Labels
Level 0
Level 1
Level 2
Level 3


File Edit Selection View Go Run Terminal Help
Search
Untitled9.ipynb
C: > Users > archa > Downloads > Untitled9.ipynb > from flask import Flask, render_template, request, jsonify
+ Code + Markdown | Run All Clear All Outputs | Outline
Select Kernel
Rainfall Distribution by Crop
Rainfall (mm)
0
50
100
150
200
250
300
rice
maize
chickpea
kidneybeans
pigeonpeas
mothbeans
mungbean
blackgram
lentil
pomegranate
banana
mango
grapes
watermelon
muskmelon
apple
orange
papaya
coconut
cotton
jute
coffee
Crop
+ Code + Markdown
Restricted Mode
Launchpad
Spaces: 4
Cell 1 of 19
CODEGPT

9. RESULT AND ANALYSIS

The Smart Farming Crop Recommendation System was successfully implemented using a Random Forest Classifier model trained on an agricultural dataset containing key features such as nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall. The model achieved a high accuracy during training and testing, demonstrating its effectiveness in predicting the most suitable crop for specific environmental and soil conditions.

When a user inputs specific soil and weather conditions into the web interface, the model predicts the most appropriate crop almost instantly. This prediction is displayed clearly on the interface. In addition to this, a chatbot was integrated into the platform to offer responses to frequently asked agricultural queries, thereby enhancing user interaction and usability.

The web interface also includes data visualization through charts showing distributions of soil nutrients and weather parameters. These visual insights help users understand patterns and correlations in the agricultural dataset. For instance, correlation heatmaps revealed that temperature and humidity were moderately correlated, which may influence crop suitability in certain regions.

Overall, the results highlight that integrating machine learning with web technologies can offer a robust and user-friendly decision support system for farmers. The model not only reduces guesswork in farming decisions but also provides a data-driven foundation to improve yield and sustainability in agriculture.

10. CONCLUSION AND FUTURE ENHANCEMENT

The Smart Farming Crop Recommendation System effectively uses machine learning to assist farmers in selecting the most suitable crops based on environmental and soil conditions. The integration of a Random Forest Classifier, a simple web interface, and a basic chatbot has made the system interactive, accurate, and user-friendly. The project has shown that data-driven farming can improve productivity and decision-making.

In the future, the system can be enhanced by integrating real-time weather data, advanced chatbot features, regional language support, and additional tools like fertilizer suggestions, pest detection, and smart irrigation. These upgrades would transform it into a more comprehensive smart agriculture assistant.